Cenwei Zhang

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Educations

The Chinese University of Hong Kong | Information Engineering *Postgraduate* Sep. 2026 — Present

Supervised by Prof. **Soung Chang Liew** and Dr. **Yuyang Du**.

Shanghai Jiao Tong University | B.Eng. *Undergraduate* Sep. 2022 — Jun. 2026

Supervised by Prof. **Lei You** (Technical University of Denmark) and Prof. **Suncheng Xiang** (SJTU).

Skills

- **Languages&Workflow**: Python; proficient in the PyTorch deep learning framework. Linux, Shell, Git, GitHub, Docker, C++.
- **Technical Foundation**: The theory of **Generative Modeling** (Diffusion & Flows) as well as manifold optimal transport mechanisms; Visual image processing model components and their variants, such as ViT, CNN, and DiT; Basic LLM/VLM frameworks, including post-traing, compression, quantization, and inference optimization methods; reinforcement Learning

Work Experience

Ubiquant-IQuest Research |*LLM algorithm intern* Apr. 2026 — Present

- Conducted research on **Best-of-Evidence** (BoE) framework that decomposes VLM candidates into verifiable evidence factors, selectively verifies high-value evidence under limited budgets, and updates candidate credibility.

DolphinAI Co., Ltd. |*Algorithm Research Intern | Medical Image Generation* Apr. 2025 — Dec. 2025

- Tackled high-frequency noise and semantic decoupling by integrating **structure-aware** strategies into the generative backbone, effectively filtering artifacts while preserving the **Manifold Consistency** of fine-grained pathological features.

Selected Publications[1]

Accepted Publication

From Baselines to Transport Geodesics: Axiomatic Attribution via Optimal Generative Flows |***NeurIPS 2026*** *Accepted*

Cenwei Zhang*, *Lin Zhu**, *Manxi Lin*, *Lei You*†

- The paper reframes feature attribution as two linked problems: allocating score change along a fixed path and selecting the path itself. It proves that, under explicit fixed-path axioms, the Aumann–Shapley line integral is the unique coordinate-wise allocation rule. It then chooses paths through minimum-kinetic-action transport from a reference distribution to the data distribution, approximated by Rectified Flow and Reflow. The theory links flow error to attribution error, while experiments support more stable and structured explanations with competitive faithfulness.

Manuscripts Under Review

Best-of-Evidence: Best-of-N Selection under Partial Verification |***ICLR 2027*** *Submission*

Cenwei Zhang, *Teng Fang*, *Yuxia Wang*, *Derek Li*, *Bryan Dai*†, *Lei You*†

- We introduce Best-of-Evidence for Best-of-N selection under partial verification. It uses a signed candidate–factor graph and an EVSI-based controller to spend a fixed budget on decision-relevant checks. It improves fixed-pool selection and rescues BoN failures when evidence is reliable and contrastive.

MedCore: Boundary-Preserving Medical Core Pruning for MedSAM |***ICLR 2027*** *Submission*

Cenwei Zhang, *Suncheng Xiang*†, *Lei You*†

- MedCore reframes MedSAM compression as medical-core extraction. It combines zeroing and reset-to-SAM interventions to preserve structures important to current prediction and medical adaptation. It then derives boundary leverage, linking contour displacement to boundary logit perturbation and local spatial slope, and turns this insight into boundary-aware Fisher pruning. The analysis shows that attention heads are more boundary-fragile than MLP structures, motivating conservative head pruning while retaining both region and contour quality.

ContourMPQ: Boundary-Localized Output Probes for Mixed-Precision Quantization of MedSAM |***AAAI 2027*** *Submission*

*Kaibing Hu**, ***Cenwei Zhang****, *Lei You*†

- We introduce ContourMPQ, a boundary-aware mixed-precision quantization framework for MedSAM. It scores deployment-matched group–bit probes in FP32 contour bands and allocates bits by exact dynamic programming. It improves region, boundary, and surface fidelity over full-grid MSE and transfers across modalities.

[1] ∗ indicates equal contribution and † indicates corresponding author

Projects

LLM Pre-training and Fine-tuning System based on nanoGPT |*Independent Project & Implementation*

Jan. 2026 — Mar. 2026

- Independently implemented the GPT-2 foundation model from scratch using PyTorch, integrating modern architectures like RoPE and the **FlashAttention** acceleration operator. Executed end-to-end pre-training on a single H100 GPU utilizing `bfloat16` mixed precision, achieving a 31% Zero-shot accuracy on the downstream HellaSwag commonsense reasoning task, outperforming the original GPT-2 baseline by approximately 10%.

Personal Summary

- Professional working proficiency in English (IELTS 7.0).
- **Personal Life**: Passionate about city walks, hiking, and amateur photography. I enjoy honing logical deduction and task planning skills through Go (inspired by AlphaGo), and often draw inspiration for constructing digital ”world models” from the open-world environment of Genshin Impact.